MICHAEL F. PÉREZ

ML & Computer Vision Engineer · Ph.D. in Computer Science · Tampa, FL (open to relocation)
michaelperez023@gmail.com | (954) 882-5069 | michaelfperez.com
linkedin.com/in/michaelperez023 | github.com/michaelperez023 | Google Scholar

ML and computer vision engineer (Ph.D. in Computer Science, University of Florida, 2026) with experience integrating computer vision into production software, building ML data and evaluation workflows, and researching video understanding across two DARPA-funded programs.

TECHNICAL SKILLS

ML & computer vision: PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, VLMs (LaViLa, LLaVA, Qwen)
MLOps & data tools: Kubernetes, Airflow, MLflow, FiftyOne, Label Studio, Docker, SLURM/HPC
Languages & systems: Python, C++, SQL, JavaScript, FastAPI, Flask, Celery, Redis, Linux, Git, NumPy, pandas, SciPy

EXPERIENCE

Senior Engineer, Raft · Tampa, FL — Aug 2026 – Nov 2026

- Developed and integrated computer vision data exploration and analysis capabilities into a larger production software system, including dataset visualization, clustering, filtering, natural-language querying, and geospatial analytics.
- Built ML data management and evaluation workflows for dataset curation, annotation review, model analysis, and debugging.
- Contributed to the development, testing, and integration of AI/ML capabilities, including object-detection experimentation and improvements to existing software systems.

Machine Learning Engineer Intern, CoVar · Durham, NC — May 2025 – Aug 2025

- Applied infrared video super-resolution to long-range object classification; first-author paper at the 2026 Military Sensing Symposium (MSS).
- Trained lightweight RGB object-detection models targeting real-time, on-device drone detection.
- Built offline computer vision evaluation pipelines spanning preprocessing, inference, metrics, and efficiency analysis.

Graduate Research Assistant, University of Florida · Gainesville, FL — Aug 2020 – Aug 2026

- Built UI and image/video analysis APIs and trained action-segmentation models for CReLeRI, a DARPA-funded explainable long-video analysis system; first author of its ACM Multimedia 2025 paper.
- Built CReLeRI's long-video analysis web tool: FastAPI backend with Celery and Redis background tasks, containerized with Docker, with GPU compute offloaded to UF's HiPerGator cluster.
- Trained and tested action-recognition models, including fine-tuning the video-language model LaViLa, for ENKIx, a DARPA-funded real-time AR task-guidance system, and ran usability and cognitive-load evaluations.
- Ph.D. thesis: developed efficient action-detection models for long videos and built a unified framework benchmarking detectors across mAP, latency, throughput, memory, and FLOPs; first-author manuscript in preparation.

EDUCATION

Ph.D. in Computer Science, University of Florida — Aug 2026
M.S. in Computer Science, University of Florida — May 2023
B.S. in Computer Science, Florida Polytechnic University — May 2020

SELECTED PUBLICATIONS

CReLeRI: Explainable, Concept-centric, Representation, Learning, Reasoning, and Interaction Video Analysis System. First author · ACM Multimedia 2025 · doi:10.1145/3746027.3754479

Super-Resolution for Improving Vehicle Recognition at Large Stand-offs. First author · MSS 2026

MuCHEx: A Multimodal Conversational Debugging Tool for Interactive Visual Exploration of Hierarchical Object Classification. Co-author · IEEE Computer Graphics and Applications, 2025 · doi:10.1109/MCG.2025.3598204

HONORS & SERVICE

Honors: L3Harris Graduate Fellowship (one of five UF CISE recipients, 2022); UF Graduate School Preeminence Award
Service: Peer reviewer for ACM Multimedia, ACM ICMR, IEEE VR, and ICXR (2024 – 2026)